

Flying Liferaft

[Aerodynamics]



[Anthony Haiser]

CID: 01719350

Group Design Project
H401 MEng Aeronautical Engineering

13/06/2022

FLR22

Subteam	Members
Technical Directors	Foster Ng
	Julian Fung
	Daniel Earl
Aerodynamics	Anthony Haiser
	Kwok Leung Yim
	Marius Borsu Lefevre
Structures	Sorawis Itthithavorn
	Antoine Rieuneau
	Olivier Sussfeld
	Jordi Frittoli
Propulsion and Systems	Matan Hamburger
	Wei Han Lee
	Natalia Melo Fernandez
	Farsai Sittichoksuwan
	Kristoff Ahlner
Performance, Stability & Control	Joshua Binns
	Elia Chimonides
	Luis Ferreira da Silva Marques
	Yi En Puah
Simulation	Bojun Vito Yang
	Gabisanth Seyon
	Rohit Singh

Abstract

Abstract. As part of the aerodynamics team on the FLR project, I undertook in the initial wing design and then the empennage design. The empennage was initially sized and then the design iterated along with input from the control systems and structures teams. The finished empennage was then analysed for aerodynamic effects due to its configuration, and interaction with a rear propeller.

Contents

List of Figures	iv
List of Tables	v
List of Symbols	vi
1 Introduction	1
2 Initial Wing Design	1
2.1 First Mission Plan	1
2.2 Design Lift coefficient	2
3 Empennage Design	2
3.1 Configuration	2
3.1.1 V-tail	2
3.1.2 T-tail	3
3.1.3 H-tail	3
3.1.4 Design Selection	3
3.2 Aerofoil Selection	4
3.3 Initial Sizing	5
3.4 Iterative process of tail design	6
3.4.1 1st Iteration	6
3.4.2 2nd iteration	7
3.5 Final Empennage Design	7
4 Empennage Analysis	8
4.1 Investigation into the 3D empennage effects	8
4.1.1 Finding the tails 3D lift coefficient	8
4.1.2 End plate effect	8
4.2 Aerodynamic effects of the tail and propeller interaction	8
4.2.1 Effect of upstream induced velocity on the tail	8
4.2.2 Effect of tail down-wash on the prop	9
5 Conclusions	11
References	11
Appendix	12

List of Figures

1	Constraint Diagram	1
2	Simplified mission plan	2
3	MQ1 Predator [4]	3
4	p.1hh hammerhead wiht a T-tail configuration [5]	3
5	Mugin Pro-5 with a H-tail configuration [6]	4
6	Initial sized tail	5
7	Initial sized H-tail	6
8	First iteration of the tail	7
9	Second iteration of the tail	7
10	Lift Coefficient versus endplate height for endplates with fixed length $l = c$ for rectangular wings with varying aspect ratios [9]	9
11	Comparison of lift coefficient of the wing prop-off and prop-wash effect versus α_{MAV} [11]	10
12	Comparison of aerodynamic efficiency of the wing prop-off and prop-wash effect versus α_{MAV} [11]	10
13	Lockheed Constellation Triple tail[13]	12

List of Tables

1	Initial sizing aerodynamic values	1
2	Initial sizing aerodynamic values	2
3	Shortlist of aerofoils to use for the tail	4
4	Initial sized tail parameters	5
5	Initial sized H-tail parameters	6
6	1st iteration tail parameters	6
7	2nd iteration tail parameters	7
8	Change in angle of attack of the horizontal tail plane due to 3D aerofoil effects . . .	8

List of Symbols

C_l	Lift coefficient
$C_{l,design}$	Design lift coefficient
$C_{l,max}$	Maximum lift coefficient
W/S	Wing loading
ρ	Air density
v	Freestream velocity
L/D	Lift to drag ratio
C_{VT}	Vertical tail volume coefficient
C_{HT}	Horizontal tail volume coefficient
L_{VT}	Vertical tail moment arm
L_{HT}	Horizontal tail moment arm
S_{HT}	Horizontal tail area
S_{VT}	Vertical tail area
b	Planform area
c	Chord length
AR	Aspect ratio
α	Angle of attack

1 Introduction

The aerodynamic design of the aircraft began after an initial sizing and baseline configuration based on the design briefs was completed. This process was using the baseline configurations of four different approaches, combined into one final design using a mediated, whole-group discussion. This presented a constraint diagram in Figure 1 to begin the wing design from as well as mission and mass requirements.

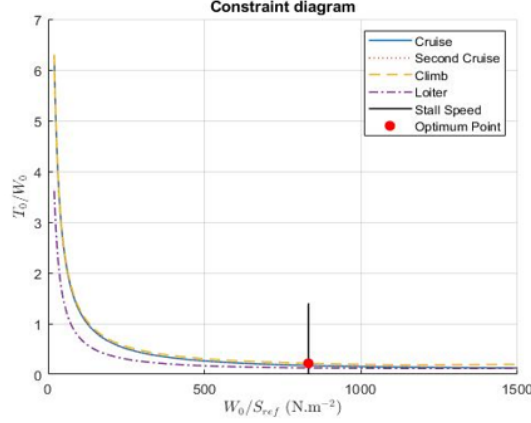


Figure 1: Constraint Diagram

These figures in Table 1 would be used as a starting point to produce initial sizes of the wing and empennage which would be the base to start iterating and optimising these components.

Table 1: Initial sizing aerodynamic values

Wing Loading	833 N/m ²
Cruise Speed	58.3 m/s
Stall Speed	38.9 m/s
Climb Rate	500 ft/min

2 Initial Wing Design

The wing design needed a starting point to build a fully optimised wing from, for the aircraft and mission requirements. So, the first aim was to design an initial wing using correction factors and the empirically designed initial sizing.

2.1 First Mission Plan

After the sizing requirements has been defined, the mission profile was fully defined so the aerodynamic components could be designed for all the flight regimes that would occur. Considering flight regimes that would be undertaken using the fixed wings primarily, a climb, first cruise segment, descent and then a second round of these regimes but for a return flight back from the mission, seen in Figure 2.

The design lift coefficient was chosen to optimise the cruise section of the flight. This was because this is a medium range search-and-deliver drone so the key part of its performance will be the ability to travel its mission distance to drop its payload. As well as this, as there is a VTOL component to the aircraft, minimising weight is key to allow the smallest batteries for this system and ensure vertical take-off can be achieved. So, designing for the longest flight regime would be the best way to ensure fuel efficiency and save weight.

To maintain the design lift-coefficient, a cruise-climb could have been employed, however, a constant altitude cruise was selected for both cruise segments. Firstly, to ensure that the aircraft

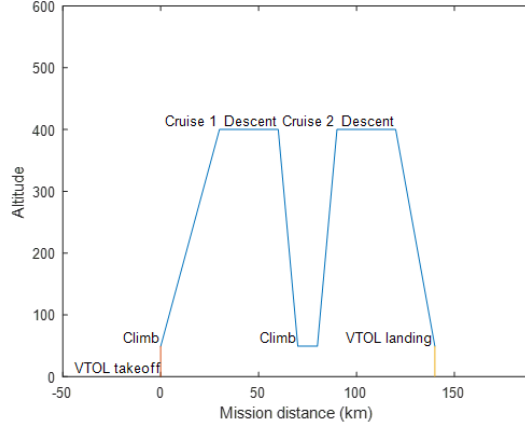


Figure 2: Simplified mission plan

did not breach the ceiling set by regulations and secondly to ensure that the control systems had a constant altitude to operate spotting systems at. Instead, angle of attack would be increased for both cruises to maintain this lift-coefficient

2.2 Design Lift coefficient

The design lift coefficient was then calculated using Equation 1. This was using the start of first cruise wing loading and cruise speed from the initial sizing, and a density based off the maximum altitude permitted by drone regulations in the area, 400ft. Some correction factors were applied to this to allow for the twist and taper which will be designed after (0.9), and the effect of other lift producing components such as the tail and fuselage (0.95) [1].

$$C_{l,design} = \frac{W/S}{\frac{1}{2}\rho v^2} \frac{1}{0.95 \times 0.9} \quad (1)$$

Whilst this lift-coefficient was known to be subject to change when the 3D effects were simulated, it allowed a good estimate of a 2D lift-coefficient from which an aerofoil could be chosen from. As well as this, the team was given an initial stall-speed of 38.9 m/s to design for, which was used to calculate a $C_{l,max}$ approximation. Both figures that would be used as a starting point for aerofoil selection are shown in Table 2

Table 2: Initial sizing aerodynamic values

$C_{l,design}$	$C_{l,max}$
0.602	1.34

3 Empennage Design

3.1 Configuration

The starting point for the design of the empennage was the configuration of the tail. To help in this choice, a literature review of similar UAVs was conducted to look at their tail configurations and the design reasons for this decision. The main UAVs that were focused on had a pusher prop configuration as this was determined to be the main driver of tail design.

3.1.1 V-tail

Notable aircraft UAV with a V-tail include the MQ-1 Predator (shown in 3 and the MQ-9 Reaper. The common advantages of a V-tail are a smaller tail area which leads to a lighter tail and less wetted area, reducing the overall drag. However, despite being more compact, the fuselage would

need to be longer to allow a greater moment arm to allow for control. Furthermore, as the Predator was initially designed for Naval use, the V-tail allows easier folding of the tail for storage on the ship [2] as well as the added military advantages of more stealth and the inverted V-tail allowing the engine to be placed above the fuselage reducing infrared signature [3]



Figure 3: MQ1 Predator [4]

3.1.2 T-tail

A tail is more controllable as it increases the moment arm of the wing to the tail without an increase in weight as well. It is also easier to recover from a spin as the elevator is located above the rudder so there is no "dead air zone" above the elevator [7]. However, during a stall, the tail is blanketed by the wing so the wing will need to be stable enough to recover from a stall without the tail if this configuration is chosen.



Figure 4: p.1hh hammerhead with a T-tail configuration [5]

3.1.3 H-tail

The main advantage of the H-tail is it offers the best position for the vertical tails to create undisturbed air. Furthermore, the vertical tails create an endplate effect [7] which will lead to a reduction in the tail area when it is further designed. Despite this, the mounting of the vertical tail at the ends of the horizontal will lead to extra structural weight. A mission specific benefit from the H-tail however comes from the lowered vertical tailplane height, which allows better storage space, as this was a design driver on the Lockheed Constellation, shown in Figure 13 in the appendix.

3.1.4 Design Selection

The V-tail design was discarded as the control system would create too much added complexity. Added fuselage length wanted to be avoided too to increase storage space and the military-based



Figure 5: Mugin Pro-5 with a H-tail configuration [6]

advantages were not relevant to our mission. The more favourable controllability and less structural weight of the T-tail meant it was chosen over the H-tail as the configuration of our design.

3.2 Aerofoil Selection

The key characteristics for choosing the tail's aerofoils were:

1. A symmetric aerofoil with a smooth lift-curve slope â this was to ensure that positive and negative lift could be created at equal angles of attack and allow smooth control input to be made.
2. A high lift to drag ratio at the design lift coefficient to minimise the drag contribution by the tail.
3. A stall angle greater than the stall angle of the wing to ensure a stable stall break.

A shortlist of 5 aerofoils was chosen based off highest lift to drag ratio whilst also meeting a suitable smooth-lift curve slope requirement, shown in Table 3

Table 3: Shortlist of aerofoils to use for the tail

Aerofoil	L/D_{cruise}	Stall Angle (Deg)	Max Thickness
NACA 0010	46.2	17.25	10%
NACA 0009	47.1	16.25	9%
NACA 66-018	38.9	19.25	18%
NACA 642-015	44.8	19.25	15%
NACA 63(3)-018	41.1	19.25	18%

The stall angle of the main wing is 15.81, hence why a selection of high stall angle aerofoils were chosen in the shortlist. The aerofoils with a 19.25 deg stall angle were chosen as there had to be a large enough safety factor for when considering the 3D effects of the tail planform on lift and stall. Therefore, the NACA 642-015 was chosen for the horizontal tail based on this high stall angle, highest lift to drag of this selection and its smaller thickness to minimise wash into the prop.

The vertical tail aerofoil was chosen as the NACA 0010 as it despite it having a slightly lower l/D , it has larger stall angle giving the aircraft more range of lateral control. Furthermore, its slightly larger thickness was favoured as the NACA 0010 and NACA 0009 were already thin aerofoils and were approaching a thickness small enough to cause extra structural problems.

3.3 Initial Sizing

At this point in the design process, stability figures such as the $C_{m,0}$ and C_g of the aircraft were not known, so empirical values based off aircraft of this configuration were used. A tail volume coefficient was selected empirically from [7] based off a single engine general aviation aircraft as this most resembled the baseline configuration â the largest configuration difference would be the pusher prop effect on tail size though this will be analysed in further iterations. Using a C_{HT} of 0.8, C_{VT} of 0.05, an initial fuselage sizing and Equation 2, Equation 3 from [7], a first tail area was calculated â shown in Table 4. This was used alongside Equation 4 to determine the aspect ratio of the wing, with a small taper ratio of 0.7 applied to both to create an initial tail planform.

$$c_{VT} = \frac{L_{VT}S_{VT}}{b_W S_W} \quad (2)$$

$$c_{HT} = \frac{L_{HT}S_{HT}}{C_W S_W} \quad (3)$$

$$AR_{tail} = \frac{2}{3} AR_{wing} \quad (4)$$

This first sizing of the tail can be shown in a to-scale plot in figure 6 compared to an estimate of the rear pusher prop, with parameters in table 4. It can be seen that, albeit an estimate, a large horizontal tail area would be needed compared to a much smaller vertical tail area. On top of this, the horizontal aerofoil has a high thickness compared to the vertical (15% versus 10% of chord). This would mean the vertical tail would have a very long and small chord for itâs wash to clear the rear prop and this long, thin structure would have to support the larger, thick and hence, heavier horizontal tail. It was decided then that this tail sizing did not support a T-tail configuration any more, and that it lends itself to a H-tail configuration.

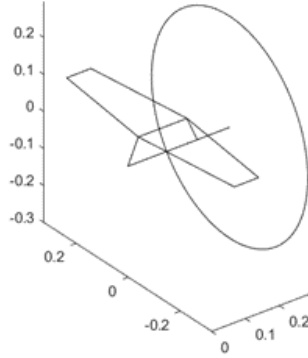


Figure 6: Initial sized tail

Table 4: Initial sized tail parameters

Section	Area	Span	Aspect ratio	Root Chord	Tip Chord
Horizontal	0.069	0.644	6	0.143	0.0715
Vertical	0.0237	0.0683	0.7867	0.2043	0.1430

Figure 7 shows the initial sizing but as a H-tail configuration. This configuration suited the initial sizing constraint a lot more as the thicker horizontal aerofoil would be supporting the smaller, thinner vertical aerofoils which is much more favourable structurally. As well as this, the larger area of the horizontal tail means it is wide enough to keep the vertical stabiliser outside of the prop and reduce wash into them. Table 5 shows the values of the initial sizing but as the new H-tail configuration.

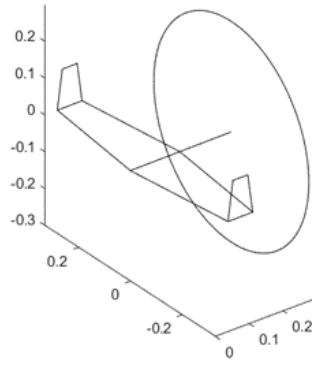


Figure 7: Initial sized H-tail

Table 5: Initial sized H-tail parameters

Section	Area	Span	Aspect ratio	Root Chord	Tip Chord
Horizontal	0.0727	0.6604	6	0.1468	0.0734
Vertical	0.0250	0.1064	3.6250	0.0734	0.0440

3.4 Iterative process of tail design

After the initial sizing produced an approximate planform, an iterative design process was set up between:

- Control systems â required planform area and size to fit control surfaces
- Structures â fuselage length creating the moment arm of the tail
- Propulsion â size of the rear pusher prop
- Aerodynamics â Aspect ratio, taper ratio and planform area

The process was set up by feeding each fuselage length change to the control systems team to determine either a change in this length or a change in planform area â this was an internal control decision. Area changes and the rear prop size were then received by the aerodynamics team where a planform was designed for each change. Once the design changes had finished, we would then have converged on a final tail design.

3.4.1 1st Iteration

The first design change came after the first control analysis was performed and need a 42.5% increase in horizontal tail area. The structural team also set a maximum tail plane span of 0.836m and the propulsion constraint remained of minimum span of 0.7m. The tail planform was then reverse engineered by calculating the aspect ratio and taper that would fit the new area and span constraint. The vertical tail aspect ratio and taper were also reverse engineered so the tip chord of the horizontal tail matches the root chord of the vertical tail to ensure a flush fit in the H-tail configuration. The updated design is shown in Figure 8 and Table 6

Table 6: 1st iteration tail parameters

Section	Area	Span	Aspect ratio	Root Chord	Tip Chord
Horizontal	0.1265	0.7324	4.2400	0.2303	0.1152
Vertical	0.0213	0.1156	2.5099	0.1152	0.0691

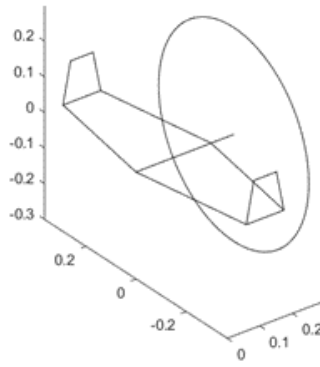


Figure 8: First iteration of the tail

3.4.2 2nd iteration

After testing in the flight simulator, it was deemed that more vertical control was required so the control systems requested a 52.8% increase in vertical tail area. This constraint, along with the first set of constraints, meant that the vertical tail needed to be redesigned. The root chord was fixed to fit to the horizontal tail and then an aspect ratio was reserve engineered from this constraint to fit the area. After this, the aspect ratio 2.48 was checked it fell in a suitable range for aerodynamic efficiency. The final values for this second iteration can be found in Table 7 and the final tail design shown in Figure 9

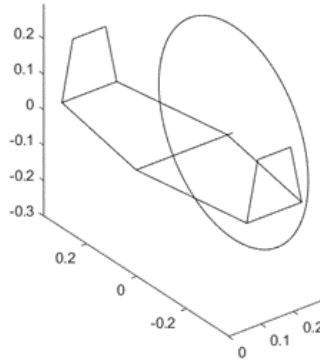


Figure 9: Second iteration of the tail

Table 7: 2nd iteration tail parameters

Section	Area	Span	Aspect ratio	Root Chord	Tip Chord
Horizontal	0.1669	0.7421	3.3000	0.2811	0.1687
Vertical	0.0452	0.1673	2.4798	0.1687	0.1012

3.5 Final Empennage Design

This iteration proved to meet the aerodynamic constraints of a suitable aspect ratio, taper and planform area. At this point the area also satisfied the control constraints and the horizontal span was large enough for the vertical fins to clear the prop diameter. This therefore meant that the teams associated with empennage had converged on a final design. From this design, no twist was

added to the planform as it not common on tail wings to allow for greater control surface area. Moreover, the Mach number of the Tail in cruise of 0.17 meant that no sweep was necessary as the tail surfaces would not encounter supersonic effects.

4 Empennage Analysis

4.1 Investigation into the 3D empennage effects

4.1.1 Finding the tails 3D lift coefficient

The final design of the tail sections was based off lift coefficients derived from the 2D aerofoil sections [8]. When applied to a 3D planform, the vortex distribution producing the wing's lift will cause the planform to have a different total lift coefficient. Previously in 2.2, this was estimated using a correction factor. However, now the planform design had been set, the total lift coefficient could be calculated in order to have an accurate figure that can be used for control design.

The analysis was performed (alongside Daniel Earl, Aerodynamics) with Athena Vortex Lattice (AVL). The program used the final planform dimensions with the NACA 642-015 for the horizontal tail aerofoil. Since the control systems team had a $C_{l,design}$ for the cruise flight regime that was necessary to maintain trim, the AVL simulation found what angle of attack the 3D aerofoil needed to be set at to meet this condition. Table 8 shows that the wing will need to be set 2.43 degrees higher in order to still make the control constraint.

Table 8: Change in angle of attack of the horizontal tail plane due to 3D aerofoil effects

Case	C_l	α
Aerofoil section	0.287	2.47
Wing	0.287	4.90

4.1.2 End plate effect

The H-tail configuration means that horizontal tail is effectively bounded by two endplates. While endplates and winglets are used to increase the efficiency of a lifting surface with a penalty of drag [9], the vertical fins are necessary for the control design. The aerodynamic side-benefits can be enjoyed as a result of this. The effect on the vortex distribution could have been simulated but instead empirical results on endplate size and its effect of the vortex distribution was used from [9]. [9] states that if the endplates are of large enough size the vortex distribution tends to a uniform one. Using empirical results, shown graphically in Figure 10 it was calculated that due to the size of the vertical tails, the horizontal tail produced a C_l 93.1% of the its sectional C_l . Despite this being based off an empirical model it showed strongly that the vertical fins were large enough to produce a near uniform vortex distribution along the horizontal tail. The previous AVL simulation was over-righted and the tail's lift coefficient was determined from it's original sectional value with the applied 93.1% correction factor. This was a more suitable method for this heavily endplated configuration.

4.2 Aerodynamic effects of the tail and propeller interaction

4.2.1 Effect of upstream induced velocity on the tail

As the empennage has a relatively unconventional configuration, with the tail and rear prop mounted so close to each other, there is likely to be some aerodynamic effects of the interaction - not covered by standard aircraft design equations. As these effects are complex and challenging to model, a literature review of the effect in similar sized aircraft at a similar Reynolds number was undertaken. Then, corrections taken from their results.

Firstly, the velocity induced upstream of the propeller will have an effect on the airflow over the horizontal tail plane (the vertical tail plane was placed outside this induced velocity). At lower Reynolds's numbers, the effect a pusher prop's upstream on a wing was very well researched.

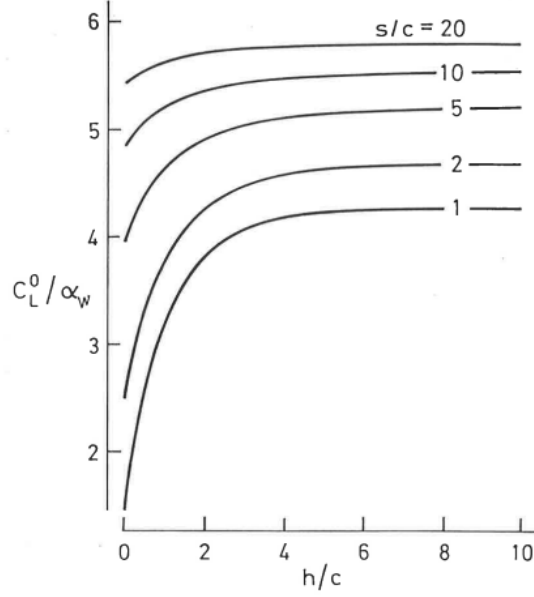


Figure 10: Lift Coefficient versus endplate height for endplates with fixed length $l = c$ for rectangular wings with varying aspect ratios [9]

Multiple papers supported the findings of [10] where the 'Fluent' CFD tool was used to solve for a pusher-prop on a micro-air vehicle (MAV). The paper found that induced upstream flow had 'been observed to retard the flow separation on the upper surface by 4 degrees and to promote the re-attachment of the separated flow'. However, the accelerated flow on the lower surface had 'been found to reduce the pressure, which contributes to the increase in pressure drag'. Although these results were gathered from Low Reynolds's flow, it did support the idea that our configuration would have added effects on the aerofoil from the propeller induced upstream velocity.

To analyse the detailed results of this effect on our configuration, [11] was looked at as it worked in the same order of magnitude as our designed Reynolds number (10^6). The experiment ran MAV's in tractor prop and pusher prop configuration and compared the two set-ups. Figure 11 shows the increase in lift coefficient across the lift curve slope of the aerofoil at an Advance ratio, J , of 0.5. This was the closest test to our cruise advance ratio of 0.833. The results from this test showed a '76% increase in the Cl_{max} ' of the wing and a delay in the stall angle.

The effect of L/D from [11] was also studied. For the same Advance ratio of 0.5, Figure 12 showed a lower L/D for the wing as the 'prop-wash effect induced a drag force to the wing'. However, in the 'Wing post-stall' region in figure 12, the efficiency of the pusher prop is comparable to a wing with no prop, due to the induced upstream velocity. This was a similar effect found in [10] where this induced velocity had helped to reattach separated flow.

The main conclusions to draw from this experiment was that the tail will produce more lift than the initial design, stall later than the initial design and ultimately have a slightly lower aerodynamic efficiency. While the higher stall angle provides negligible benefit as the wing is already designed to stall beforehand, it provides an added safety factor that the tail stalls afterwards and ensures a stable stall break. The added Cl_{max} of the horizontal tail will impact the static margins of the aircraft. The ideal way to model this increase would be a wind tunnel test to see the exact the Cl_{max} change for the designed planform. Then, by decreasing the tail's angle of attack, the true Cl can be matched to the Cl which was designed. This is outside the level of design needed for the time span, but will be a necessary test if the aircraft is due to be fully produced.

4.2.2 Effect of tail down-wash on the prop

After investigating the effect of prop induced velocity on the tail, research was conducted into whether the down-wash produced from the horizontal tail would have a significant impact on the

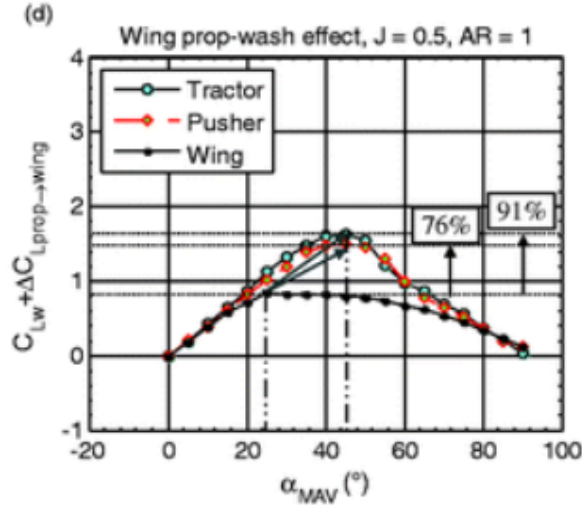


Figure 11: Comparison of lift coefficient of the wing prop-off and prop-wash effect versus α_{MAV} [11]

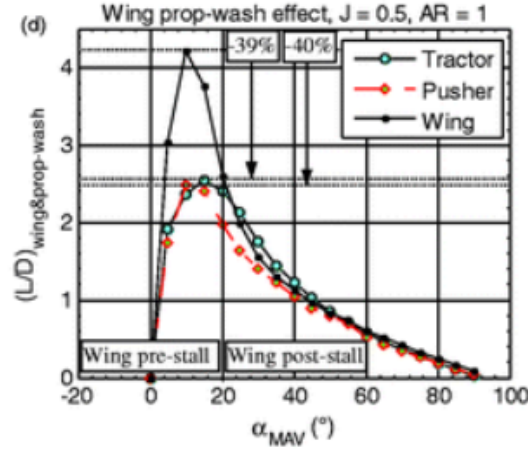


Figure 12: Comparison of aerodynamic efficiency of the wing prop-off and prop-wash effect versus α_{MAV} [11]

pusher prop's performance. A NASA study ([12]) on the impact of tail wash into a motor identified that tail wash could cause an 'unsteady bladeloading caused by the blade interactions with the wake mean velocity distribution [and have a] strong effect on the harmonics of blade-passage noise'. The test was at a Mach number of 0.13 which is comparable to the Mach number of the tail of 0.17. The study identified that if the propeller was more than one chord length behind the empennage, this effect became weak. The final mean aerodynamic chord of the horizontal tail was 0.225m, so it was advised that the prop be placed this distance from the rear of the tail. However, the structures team were constrained by 0.04m length and this was deemed of higher importance to the design in order to meet the storage requirements. Due to this decision, the prop would placed be at 0.178 of the horizontal tail's chord and would be susceptible to unsteady blade loading and 'blade passage harmonics'. To investigate this loading, an aerodynamics and acoustic study in the form of [12] would need to be conducted. If the aircraft is due to be fully produced, it would need this further wind tunnel testing to ensure that rear prop is not experiencing these harmonic loads in its key flight regimes.

5 Conclusions

The key method of design used for this project was initial sizing, then iterations converging onto a final design. This worked well as the baseline configuration was mostly standard so many empirical models were available to produce an accurate initial sizing result. The process of iterating the design between groups worked well when clear constraints were defined by each group. This allowed changes to be made in each iteration outside these constraint.

The final empennage design is limited in some aerodynamic aspects as the design was constraints by more important requirements of other groups. Ideally, a wider span horizontal tail would be used as this would increase the tail's aspect ratio and make this wing more aerodynamically efficient. However, this was unable to be met due to the structural constraint of a maximum tail span. Furthermore, a whole chord length of distance between the end of the tail and the prop would be ideal for avoiding the tail's wash but, once again, a more important fuselage length constraint took priority.

As stated in the analysis, as the empennage has a slightly complex configuration (being a H-tail with a closely mounted propeller), more in depth analysis of the interactions are needed to fully estimate its performance. If the project allowed more time and resources, a full model of the empennage and rotating propeller in Star-CCM+ would be advisable to properly see how the propeller interaction is affecting the flow on the tail surfaces. Furthermore, a wind tunnel test of the full size model is achievable due to its compact wingspan. This would be necessary before the actual mission to find the tail's exact lift coefficient as a culmination of all the surface effects. This can also be used to check if the rear propeller is experiencing any harmonic forcing from the tail wash, in any of the key flight regimes. These tests would of course be to ensure complete detail in the empennage design - the methods used in this report still provide a very accurate approximation of a complete empennage design and the effects upon it.

References

- [1] Torenbeek, E. (1982) *Synthesis of Subsonic Airplane Design : An Introduction to the Preliminary Design of Subsonic General Aviation and Transport Aircraft, with Emphasis on Layout, Aerodynamic Design, Propulsion and Performance*. Springer Netherlands.
- [2] Kampf, P. (2015) Why are military drones shaped so strangely? <https://aviation.stackexchange.com/questions/21766/why-are-military-drones-shaped-so-strangely> [Accessed 12 June 2022].
- [3] Pros and cons of V-tail. (2019) <https://prosancons.com/aircraft/pros-and-cons-of-v-tail/> [Accessed June 14 2022].
- [4] Panzino, C. (2017) New in 2018: Air Force will officially retire MQ-1 Predator drone. <https://www.defensenews.com/air/2017/12/30/new-in-2018-air-force-will-officially-retire-mq-1-predator-drone/> [Accessed June 14 2022].
- [5] Piaggio Aerospace. <http://www.p1hh.piaggioaerospace.it/> [Accessed June 14 2022].
- [6] Mugin VTOL UAV AIRFRAMES. <https://www.muginuav.com/>.
- [7] Raymer, D. (1989) *Aircraft Design: A Conceptual Approach* . Sixth Edition.
- [8] Airfoil Tools. <http://airfoiltools.com/> [Accessed June 14 2022].
- [9] Standingford, D. (1997) *Optimal Lifting Surfaces*. The University of Adelaide.
- [10] Choi, S. Ahn, J. (2010) A computational study on the aerodynamic influence of a pusher propeller on a MAV. 40th AIAA Fluid Dynamics Conference. , American Institute of Aeronautics and Astronautics Inc.
- [11] Chinwicharnam, K. Thipyopas, C. (2016) Comparison of wingâpropeller interaction in tractor and pusher configuration. *International Journal of Micro Air Vehicles*. <https://journals.sagepub.com/doi/full/10.1177/1756829316638206>.
- [12] Soderman, P. Horne, W. (1990) *Acoustic and aerodynamic study of a pusher-propeller aircraft model*.

- [13] Johnson, J. L. (2013) FLIGHT OF THE CONNIE: THE INCREDIBLE STORY OF THE STAR OF AMERICA. <https://www.airlinereporter.com/2013/11/flight-of-the-connie-the-incredible-story-of-the-star-of-america/> [Accessed June 14 2022].

Appendix



Figure 13: Lockheed Constellation Triple tail[13]